

SKAGWAY, AK, USA

WMO: 703620

Lat: 59.467N

Lon: 135.300W

Elev: 20

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 25335

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	6.8	-17.8	2.1	4.2	-13.2	2.7	8.4	40.4	23.7	37.5	26.0	24.1	30	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	72.2	57.7	68.4	56.6	65.4	55.5	59.5	68.3	58.2	65.7	57.0	63.4	10.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	65.5	61.1	54.5	63.2	60.1	53.6	61.2	59.3	26.0	68.3	25.1	65.7	24.3	63.5	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.9	30.0	26.6	1.0	80.4	5.4	5.4	-2.8	84.3	-6.0	87.5	-9.0	90.5	-12.9	94.5		
			-0.8	62.3	5.0	4.3	-4.4	65.4	-7.3	67.9	-10.1	70.3	-13.8	73.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	27.6	30.1	33.1	41.6	50.2	55.9	57.9	57.0	51.1	43.4	33.6	29.4
	DBStd	13.00	11.49	8.85	8.04	4.93	4.96	4.08	3.48	3.77	3.91	4.80	8.09	8.93
	HDD50	3467	696	559	523	257	59	1	0	1	31	209	492	639
	HDD65	8173	1160	979	988	701	460	277	223	251	417	671	942	1104
	CDD50	774	0	0	0	5	64	177	244	217	64	3	0	0
	CDD65	8	0	0	0	0	0	3	3	2	0	0	0	0
	CDH74	77	0	0	0	0	3	27	19	28	0	0	0	0
	CDH80	14	0	0	0	0	0	3	4	7	0	0	0	0
	WSAvg	10.4	14.2	12.1	11.7	9.4	9.9	10.2	9.4	8.3	8.3	9.1	10.3	12.0
	PrecAvg	38.5	3.6	3.0	2.6	2.0	1.5	1.3	1.7	2.6	5.1	6.3	4.5	4.1
Precipitation	PrecMax	43.6	7.9	5.6	8.4	4.4	3.5	3.2	5.2	5.6	8.0	11.5	11.5	9.1
	PrecMin	29.9	1.1	0.6	0.3	0.6	0.5	0.4	0.2	0.5	2.7	3.5	0.4	1.1
	PrecStd	3.8	1.8	1.3	1.7	1.0	0.8	0.6	1.1	1.3	1.4	1.9	2.2	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	46.3	53.6	66.6	72.4	77.8	76.0	77.4	68.3	56.0	49.3	48.7
		MCWB	43.6	40.4	43.0	49.4	53.4	58.9	60.1	56.8	53.5	47.8	44.3	44.2
	2%	DB	44.9	44.2	48.2	58.1	67.2	71.2	70.3	70.2	62.7	53.5	46.7	44.9
		MCWB	41.0	39.2	40.5	44.3	51.8	58.4	59.1	58.0	52.4	46.9	42.3	41.3
	5%	DB	42.7	42.5	45.0	53.6	63.2	66.6	66.6	66.1	59.4	51.8	44.9	42.7
		MCWB	39.0	38.2	38.9	42.5	50.8	56.1	57.7	56.7	52.0	46.5	41.0	39.1
	10%	DB	40.5	40.3	42.6	49.7	59.4	63.2	63.8	63.3	57.0	50.0	43.2	40.4
		MCWB	37.2	36.6	37.7	41.9	49.4	54.5	56.5	55.9	51.4	45.2	39.7	37.4
	0.4%	WB	44.4	42.1	45.3	50.2	55.7	62.2	61.6	60.6	57.0	51.0	45.8	44.6
		MCDB	48.4	45.3	51.4	64.9	66.2	71.0	71.8	71.5	62.7	53.8	48.4	47.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.3	40.4	41.9	46.3	53.5	58.8	59.7	59.0	54.9	49.3	43.5	41.7
		MCDB	44.5	43.3	46.6	55.0	64.7	69.0	68.0	67.0	59.8	52.0	45.8	45.0
	5%	WB	39.4	38.8	39.9	44.2	51.7	56.8	58.3	57.8	53.4	47.6	41.7	39.4
		MCDB	42.4	41.5	44.0	51.0	60.7	65.4	65.5	64.3	57.4	50.5	44.3	42.0
	10%	WB	37.6	37.2	38.3	42.6	50.3	55.0	57.0	56.7	52.2	46.1	40.2	37.6
		MCDB	40.0	39.7	41.8	48.6	58.3	62.2	63.1	62.3	56.0	49.0	42.7	39.9
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.6	10.0	12.9	13.7	12.2	10.5	10.7	9.6	8.5	7.0	7.1
		MCDBR	10.1	9.8	14.1	21.6	22.5	21.3	17.8	18.3	14.9	10.4	8.0	9.5
		MCWBR	7.4	6.7	8.4	10.9	10.7	10.2	8.3	8.3	7.2	5.9	5.7	7.1
	5% WB	MCDBR	9.3	8.7	11.9	16.9	18.6	18.7	15.7	14.9	11.2	8.3	7.6	9.3
		MCWBR	7.2	6.6	7.9	9.5	9.4	9.5	7.8	7.5	6.4	5.8	5.7	7.2
Clear-Sky Solar Irradiance	taub		0.288	0.300	0.332	0.374	0.386	0.423	0.413	0.392	0.350	0.312	0.272	0.253
	taud		1.978	2.078	2.013	1.947	2.045	2.068	2.173	2.248	2.315	2.187	2.009	1.943
	Ebn at Noon		142	217	247	257	265	256	256	252	244	211	148	110
	Edh at Noon		17	26	38	48	47	47	42	36	28	23	16	12
All-Sky Solar Radiation	RadAvg		124	362	784	1306	1742	1653	1496	1140	734	344	158	77
	RadStd		10	52	81	128	148	104	116	114	72	40	18	6

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+3.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	+2.27	+3.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page